



Maritime &
Coastguard
Agency

Guidance

MGN 641 Amendment 1

Published 9 March 2022

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1. Introduction

1.1 Chapter IV of the Annex to the International Maritime Organization's (IMO) International Convention for the Safety of Life at Sea, 1974 ("SOLAS") makes provision for radio communications for ships to which the Convention applies.

1.2 The Merchant Shipping (Radiocommunications) (Amendment) Regulations 2021 ("the 2021 Regulations") give effect to the requirements of Chapter IV of the Annex to SOLAS. Future amendments to Chapter IV will be given legal effect in the UK by means of the ambulatory reference provision in regulation 7. This provision ensures that when amendments come into force internationally, they will at the same time take effect in UK law.

1.3 This Note refers to ships to which Chapter IV applies, in accordance with regulation 4 of the 2021 Regulations; unless there is express provision to the contrary, this Note will apply to all such ships. As the 2021 Regulations refer the reader directly to Chapter IV, this Note is intended to compliment the 2021 Regulations by clarifying aspects of the Chapter IV requirements; only reference to those requirements that need further explanation will be found here.

2. Recognised satellite service provider – Regulation 2 of Chapter IV of SOLAS

2.1. All UK ships should note for the purpose of regulatory compliance that when choosing a recognised mobile satellite service (as defined in Regulation 2.1.17 of Chapter IV of SOLAS) that the service should:

2.1.1. be recognised by the IMO;

2.1.2. be the subject of a letter of compliance from International Mobile Satellite Organisation (IMSO); and

2.1.3. have achieved Full Operational Capability (FOC) for the required services.

2.2 A non-UK ship may use satellite service providers that have been recognised by their flag state which fulfil the relevant functional requirements as outlined in Chapter IV of SOLAS.

3. Radio Installation – Regulation 6 of Chapter IV

3.1 Installations on UK ships should at least follow the IMO COMSAR Circ.32 (as amended) 'Harmonisation of GMDSS requirements for radio installations on board SOLAS ships. The harmonised guidelines give an unambiguous interpretation of the radio installation requirements in Chapter IV of SOLAS.

3.2 Regulation 6.2.5 of Chapter IV refers to the marking of radio installations; consideration should be given to times of distress or urgent communications where relevant ships identities would be required. This could include, but not be limited to, EPIRB hex identification, AIS SART, MMSI, ship IMO number and satellite communication numbers.

3.3 Mandatory handheld GMDSS VHF can be used to fulfil the requirements for handheld VHF transceivers and communications from the wings of the navigation bridge as mentioned in Regulation 6.3 of Chapter IV and COM/Circ.105. Alternatively, a simplex VHF transceiver (single frequency only) or a remote-controlled unit with optional channel selector, loudspeaker and microphone may be installed in these positions. Remote-controlled units with no channel selector will have a channel selector convenient to the conning position.

4. Sources of energy – Regulation 13 of Chapter IV

4.1 For the purpose of calculating the required capacity of the reserve source of energy, the total current used in calculations should be equal to the highest sum of all the radio installations which simultaneously can be connected to the source of energy, based on the following –

- 4.1.1. the current consumption of the VHF receiver;
 - 4.1.2. one fifth of the current consumption of the VHF transmitter;
 - 4.1.3. the current consumption of a MF or MF/HF receiver and of the transmitter when it is in condition that operation of the “press to transmit” switch will make it ready for immediate transmission;
 - 4.1.4. one third of the current which may be drawn by a MF or MF/HF transmitter for speech transmission on the frequency at which the current consumption of the transmitter is at a maximum;
 - 4.1.5. the current consumption of a ship earth station when it is receiving transmissions;
 - 4.1.6. one quarter of the current which may be drawn by a ship earth station when it is transmitting in the mode at which the current consumption is at a maximum; and
 - 4.1.7. the total current consumption of all additional loads to which the reserve source may supply energy in times of distress or emergency.
- 4.2 The following formula is recommended for determining the electrical load to be supplied by the reserve source of energy for each radio installation required for distress conditions: $\frac{1}{2}$ of the current consumption necessary for transmission + the current consumption necessary for reception + the current consumption of any additional loads.
- 4.3 An appropriate method of checking the capacity of an accumulator battery is to fully discharge and recharge the battery, using normal operating current and period (e.g. 10 h). Assessment of the charge condition can be made at any time, but it should be done without significant discharge of the battery when the ship is at sea.
- 4.4 Guidelines on the configuration of the reserve source or sources of energy used to supply radio installations on GMDSS ships are found in COMSAR/Circ.16.

5. Performance Standards – Regulation 14 of Chapter IV

5.1 On the basis of paragraph 3.2 of MSN 1874 (M+F) Amendment 5, all existing approved radio installations on board a UK ship will continue to be

accepted providing the equipment continues to operate in accordance with the existing approval.

5.2 VHF radiocommunication equipment should comply with IMO resolution A.803(19) on shipborne VHF radio installations and IMO MSC.1/Circ.1460 (as amended) on the validity of shipborne radio equipment. Clarification on the amended channelisation required of VHF radio equipment can be found in MGN 562 (as amended).

6. Maintenance Requirements – Regulation 15 of Chapter IV

6.1 For the purposes of Regulation 15.5 of Chapter IV of SOLAS, there are three methods of ensuring availability of GMDSS radio installations:

6.1.1. At sea electronic maintenance;

6.1.2. Shore-based maintenance; and

6.1.3. Duplication.

6.2 Maintenance of a ships radio installation can be carried out by a qualified person at sea, by shore-based maintenance or a mixture of both. IMO Resolution A.702(17) gives guidelines for GMDSS radio maintenance in sea areas A3 and A4. All ships that rely upon at-sea maintenance must carry adequate information, tools and spares for equipment to be properly operated and maintained while the ship is at sea.

6.3 At sea maintenance of radio communication installations is to be carried out by qualified personnel such as an Electro-Technical Officer (ETO) or other appropriate qualifications that meets the requirements of Chapter III/6 STCW 1978, such as a GMDSS radio maintenance course.

6.4 If availability is ensured by using shore-based maintenance as part of a combination of methods, an arrangement acceptable to the MCA should be established to ensure adequate support of the ship for the maintenance and repair of its radio installations.

6.5 Acceptable arrangements are –

6.5.1. A signed declaration from the owner or his representatives setting out the arrangement with a company or companies for covering the

trading area of the ship to provide maintenance and repair facilities on a call-out basis;

6.5.2. Provision of facilities at the main base of ships engaged on a regular trading pattern.

6.6 Duplication ensures availability in case of failure of one of the radio installations while the ship is at sea. Duplication would mean an uninterrupted service ensuring compliance with Regulation 4 of Chapter IV of SOLAS.

6.7 For the purposes of Regulation 15.3 of Chapter IV, the guidelines contained in clause 1.4 and 1.5 of IMO COMSAR/Circ.32 (Harmonization of GMDSS Requirements for Radio Installations on Board SOLAS Ships) describe a means of providing adequate information.

6.8 The EPIRB annual test required by Regulation 15.9.1 of Chapter IV should be conducted in accordance with IMO MSC/Circ 1040/Rev.2 (Guidelines on annual testing of 406 MHz Emergency Position-Indicating Radio Beacons (EPIRBs)).

6.9 The UK Administration accepts organisations nominated and approved by the EPIRB manufacturer to carry out shore-based maintenance required by Regulation 15.9.2 of Chapter IV. Such maintenance should be performed in accordance with IMO MSC/Circ 1039/Rev.1 (Guidelines for shore-based maintenance of Emergency Position-Indicating Radio Beacons (EPIRBs)).

7. Radio Personnel – Regulation 16 of Chapter IV

7.1 In accordance with MSN 1864(M) and the STCW Code, all personnel designated to perform functions of a GMDSS radio operator on board UK ships must be qualified. The MCA currently issues the following certificates through the Association of Marine Electronic and Radio Colleges (AMERC):

7.1.1. General operator's certificate (GOC);

7.1.2. Restricted operator's certificate (ROC).

7.2 An officer or member of crew holding an equivalent radio operator's certificate that is recognised by the MCA will be deemed qualified; this

includes First-class radio electronic certificate or Second-class radio electronic certificate (which the MCA does not itself issue).

8. Radio Records - Regulation 17 of Chapter IV

8.1 The MCA recommends that completed logs are retained on board ship for a minimum period of at least one year, and that after this period completed logs are kept for a total period of seven years either on board or ashore.

8.2 MGN 530 (M+F) applies for the purposes of the information in respect of purchasing of the MCA official GMDSS radio log.

8.3 The MCA recommends that the GMDSS radio log include the ship name, call-sign and MMSI and should be completed in accordance with the column headings and include:

8.3.1. Radio equipment checks including batteries, comments, and date performed;

8.3.2. Distress, Urgency and Safety traffic;

8.3.3. Operator on and off watch;

8.3.4. Record agreement by Master;

8.3.5. For cargo ships, the operator designated primary responsibility for communications during distress incidents is as required by SOLAS IV/16.1

8.3.6. For passenger ships, the operator assigned to radiocommunications duties only during distress incidents is as required by SOLAS IV/16.2.

More information

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